

Problem 11.26

Amin deposits 10,000 in a bank. During the first year the bank credits an annual effective rate of interest i . During the second year the bank credits an annual effective rate of interest $(i - 5\%)$. At the end of two years, Amin has 12,093.75 in the bank. What would Amin have in the bank at the end of three years if the annual effective rate of interest were $(i + 9\%)$ for each of the three years?

Solution

First, use the actual two-year accumulation to determine i :

$$\begin{aligned}10,000(1 + i)(1 + i - 0.05) &= 12,093.75, \\(1 + i)(0.95 + i) &= 1.209375.\end{aligned}$$

The rates 12.5% in the first year and 7.5% in the second year satisfy this equation, since

$$(1.125)(1.075) = 1.209375.$$

Therefore, $i = 0.125 = 12.5\%$, and the hypothetical annual effective rate for each year is

$$i + 9\% = 12.5\% + 9\% = 21.5\%.$$

Starting from the original deposit of 10,000, the amount after three years at this rate is

$$\begin{aligned}10,000(1 + 0.215)^3 &= 10,000(1.215)^3 \\&\approx 17,936.13.\end{aligned}$$

Thus, Amin would have

$$\boxed{17,936.13}$$

at the end of three years under the stated hypothetical rate.